MILITARY SPECIFICATION

CLOTH, TWILL, ARAMID, STAPLE, HIGH TEMPERATURE RESISTANT

This specification is approved for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope This specification covers high temperature resistant aramid twill cloth and is intended for use in the manufacture of aeronautical clothing

1.2 Classification. The cloth shall be of the following types, as specified (see 6 2).

Type I	3.45 oz. per square yard
Type II	5.0 oz. per square yard

2. APPLICABLE DOCUMENTS

2 1 Government documents.

* 2.1.1 Specifications and standards The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Federal

V-T-295	Thread, Nylon
PPP-P-1133	Packaging of Synthetic Fiber Fabrics

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to. Systems Engineering and Standardization Department (Code 53), Naval Air Engineering Center, Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

Federal

FED-STD-191	Textile Test Methods

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

* 2 1 2 Other Government documents. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation

FEDERAL TRADE COMMISSION

Textile Fiber Products Identification Act

*(Copies of specifications, standards, handbooks, drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

* 2.2 Other publications. The following document forms a part of this specification to the extent specified herein Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS

AATCC Technical Manual

(Applications for copies should be addressed to the American Association of Textile Chemists and Colorists, P O. Box 12215, Research Triangle Park, NC 27709)

AMERICAN SOCIETY OF TESTING AND MATERIALS

ASTM D 1577	Test Method for Linear Density of Textile Fibers

(Applications for copies should be addressed to the American Society of Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

COLOR ASSOCIATION OF THE UNITED STATES, INC

TCA Cable Numbers and Colors

(Applications for copies should be addressed to the Color Association of the United States, Inc., 200 Madison Avenue, New York, New York 10016)

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*(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

* 2 3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained

3 REQUIREMENTS

3.1 Materials. The cloth shall be made from a high strength, non-melting aramid staple yarn.

3.2 Weave The weave for Types I and II shall be a herringbone twill as shown in Figure 1.

3.3 Color Unless otherwise specified, the color of the cloth shall be green or orange. The green shall match Navy Shade No. 3434 or Air Force Sage Green 1565 and the orange shall be Indian Orange TCA Cable No. 70072.

3.3.1 Colorfastness. The dyed finished cloth shall show fastness to light and laundering equal to or better than the standard sample. When no standard sample has been established or designated as applicable to colorfastness, the dyed and finished cloth shall show "fair" fastness to light at 20 Standard Fading Hours and "good" fastness to laundering when tested as specified in 4.3.

3 4 Physical requirements. The finished cloth shall conform to the requirements shown in Table I when tested as specified in 4.3

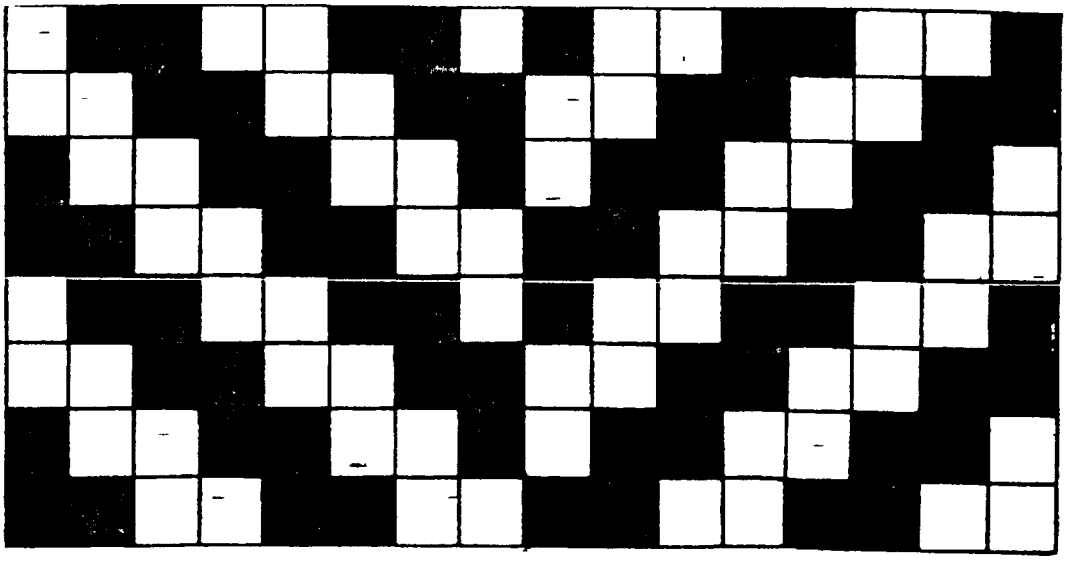
3.5 Finishing The cloth shall be desized and scoured The dyed and finished cloth shall be given a durable antistatic treatment (see 4.3) so that the maximum resistivity before laundering shall be 3 0 x 10^{11} ohms per square and the maximum resistivity after five (5) launderings shall be 8 0 x 10^{11} ohms per square when tested as specified in Table VI The cloth shall be set by autoclaving (see 6.4)

3.5 1 Non-fibrous material The total chloroform-soluble and water-soluble material content of the finished Types I and II cloth after application of the anti-static treatment shall be not less than 0.4 nor more than 2 0 percent when tested as specified in Table VI

3.6 Curling When tested as specified in 4.3.1, the finished cloth shall lie flat, without distortion, and show no evidence of curling

3.7 Width The width of the cloth shall be as specified (see 6 2), and shall be the minimum acceptable width inclusive of selvage when fly shuttle looms or shuttleless with tuck-in selvage looms are used For all other shuttleless looms, the width measurement shall be made between the last warp yarn on each side excluding the protruding fringe

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WEAVE REPEATS ON 16 ENDS AND 4 PICKS

Figure 1 Weave design

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3 8 Length and put-up Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 40 yards The lengths shall be put up on rolls as specified in PPP-P-1133.

TABLE I. Physical requirements.

Characteristics	Requirements	
	Type I	Type II
Staple denier	1.5	1 5
Staple length, inch (nominal)	1.5 to 2	1 5 to 2
Yarns per inch, min		
Warp	107	94
Filling	75	76
Weight, ozs per sq yd	3 5 -0 05 +0.3	5.0 - 0 +0 5
Breaking strength, lbs, min.		
Warp	90	130
Filling	60	100
Tearing strength, lbs, min		
Warp	6 0	16 0
Filling	5 0	10.0
Air permeability, cu ft air/min ft^2 at 1/2 inch water pressure, min.	200	100
Shrinkage in laundering, percent, max		
Warp	2 0	2.0
Filling	1 5	2 0
Sewability, seam efficiency, percent, min.	80	80
Flame resistance		
Flaming time, seconds, max	2	2
Glow time, seconds, max.	25	25
Char length, inches max., average	3 5	2 5
Acidity (pH)	5 5-8 5	5 5-8 5

3 9 Fiber identification Each roll of cloth shall be labeled, ticketed, or invoiced for fiber content in accordance with the Textile Fiber Products Identification Act.

3.10 Workmanship The quality of manufacture shall be controlled to guarantee conformity to the high standards of workmanship required by the acceptable quality levels specified in Section 4.

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4 QUALITY ASSURANCE PROVISIONS

4 1 Responsibility for inspection Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements

* 4.1.1 Responsibility for compliance All items must meet all requirements of Sections 3 and 5 The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Sampling and inspection for acceptance Sampling and inspection shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated

4 2 1 Inspection of the end item. The end item shall be inspected for defects in accordance with classifications in 4.2.1.1, 4.2.1.2 and 4 2 1.3

4.2 1 1 Yard-by-yard examination. The required yardage of each roll shall be examined and the visual defects classified as listed in Table II. All defects found shall be counted regardless of their proximity, one to another, except where two or more defects represent a single local condition of the cloth, in which case only one defect shall be counted for each warpwise yard or fraction thereof in which it occurs. The unit of product for this examination shall be 1 linear yard The sample size shall be in accordance with Inspection Level II of MIL-STD-105 The acceptable quality level shall be 6 5 defects per 100 units (yards). The lot size shall be expressed in units of one (1) yard each An approximately equal number of yards shall be examined from each roll selected The number of rolls from which the sample is to be selected shall be in accordance with Table III.

4 2 1 2 Overall examination. Each defect listed in Table IV shall be counted not more than once in each roll examined. The unit of product shall be one roll The sample size or number of rolls selected as samples and the maximum number of defects acceptable, shall be as shown in Table III.

4 2 1 3 Examination for length

4.2 1.3 1 Individual rolls During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than two yards less than the length marked on the ticket shall be considered a defect with respect to length The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length

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TABLE II. List of defects

Broken or missing yards - two contiguous, 2 inches or more in length
Reed mark - clearly visible.
Floating warp ends or filling clearly visible.
Hole, cut, tear - any
Spot, stain or streak more than 2 inches in combined directions
Selvage, cut, torn, tight, rolled or loopy
Open, thin, abraded or weak place.
Foreign or waste fiber, more than 1 inch in combined directions

TABLE III. Sampling for test

Lot Size in Yards	Sample Size in Rolls	Acceptance Number
Up to 1300	3	0
1301 to 3200	5	0
3201 to 8000	7	0
8001 to 22,000	10	0
22,001 to 110,000	15	0
110,001 and up	25	1

TABLE IV. Overall defects

Overall uncleanness, objectionable odor
Uneven weaving for more than 20 yards 1/
Selvage curled, folded, rolled, slack, or loopy, over more than half the piece 1/
Baggy, ridgy or wavy cloth
Width, less than minimum specified

1/ Intermittently or continuously

4.2.1.3.2 Total yardage in sample The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the ticket

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4 2 2 Examination of preparation for delivery requirements. An examination shall be made in accordance with the provisions of PPP-P-1133 to determine compliance with the preservation, packing and marking requirements of Section 5 of this specification.

4 2 3 Examination for compliance with the Rules and Regulations under the Textile Fiber Products Identification Act. During the yard-by-yard examination, each roll in the sample shall be examined for conformance to the Rules and Regulations under the Textile Fiber Products Identification Act. Each roll not labeled in accordance with this Act shall be a defect The lot shall be unacceptable if two or more of these defects occur

4 3 Testing of the end item The methods of testing specified in FED-STD-191, wherever applicable, and as listed in Table VI shall be followed The physical and chemical values specified in Section 3 apply to the average of the determinations made on a unit of product for test purposes as specified in the applicable test method. The lot size shall be expressed in units of 1 linear yard. The unit of product for test purposes shall be 5 linear yards, full width of the finished cloth The sample size shall be as listed in Table V The lot shall be unacceptable if any unit fails to meet any requirement specified

4 3.1 Curling. A specimen of cloth 1-1/2 by 6 inches long shall be cut in the warp and filling directions and placed on a flat surface for at least 5 minutes and then visually examined for evidence of curling The specimen shall be taken no less than 6 (six) inches from the selvage.

TABLE V Sample size

Lot Size (Yards)	Sample Size
800 or less	2
801 to 22,000	3
22,001 and over	5

TABLE VI. Test methods

Characteristic	Reference Paragraph	Test Method
Material Identification	3.1	
Weave	3 2	Visual
Finish	3 5	
Staple Denier	Table I	ASTM D 1577
Staple Length	Table I	Visual

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TABLE VI Test methods (continued)

Characteristic	Reference Paragraph	Test Method
Yarns per inch		
Warp	Table I	5050
Filling	Table I	5050
Weight	Table I	5041
Breaking Strength, Types I & II		
Warp	Table I	5104
Filling	Table I	5104
Tearing Strength, Types I & II		
Warp	Table I	5134
Filling	Table I	5134
Air Permeability	Table I	5450
Shrinkage in Laundering, Types I & II 3/		
Warp	Table I	5556
Filling	Table I	5556
Sewability 1/	Table I	5110
Flame Resistance 2/	Table I	5903
Total Chloroform-Soluble and Water Soluble Material	3.5.1	2611
Curling	3 6	4 3.1
Acidity (pH)	Table I	2811
Colorfastness to		
Light	3 3 1	5660
Laundering	3 3.1	5610
Antistatic finish		
Before laundering	3.5	4/ 5/
After five (5) launderings	3 5	5556 3/ 4/ 5/

1/ For Types I and II the needle size shall be 16, and the thread shall conform to V-T-295, Type II, Class 1, Size B.
2/ Warp direction only.
3/ Use cotton laundering schedule, without sour
4/ AATCC Test Number 76-1978, Electrical Resistivity Fabrics
5/ Average of three determinations to the nearest 1 0 x 10" ohms per square

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5 PACKAGING

* 5 1 Preservation Preservation shall be Level A or C as specified (see 6.2).

* 5 1.1 Levels A and C. The cloth shall be preserved in accordance with the applicable requirements of PPP-P-1133

5.2 Packing Packing shall be Level A, B or C as specified (see 6.2)

5 2 1 Levels A, B and C The cloth shall be packed in accordance with the applicable requirements of PPP-P-1133.

5.3 Marking. In addition to any special marking required by the contract or order, shipment shall be marked in accordance with the requirements of PPP-P-1133

6 NOTES

6 1 Intended use. This cloth is intended for use in the fabrication of lightweight flight clothing

6 2 Ordering data Procurement documents should specify the following:

a Title, number and date of this specification.

b. Color (see 3.3).

c Quantity and dimensions of cloth required (see 3 7 and 3.8)

d Selection of applicable levels of preservation and packing (see 5 1 and 5 2).

e Type required (see 1 2)

6 3 Types I and II cloth The cloth has been found to meet the specification requirements when woven using yarn counts (cotton system) of 35's to 38's for Type I and 22's and 24's for Type II with a twist multiplier of 4 50 minimum and finished without calendering The cloth has been found to meet the specification requirements when woven using yarn which was twist set, under pressure, prior to winding.

6 4 Autoclaving. One suggested method for autoclaving is as follows:

At the start of autoclaving and as soon as the specified pressure is attained, the steam shall be drawn through the cloth by the application of a vacuum for a period of five (5) minutes in order to adequately saturate the cloth with steam The pressure shall be increased to the thirty psi minimum for approximately twenty minutes The steam shall then be cut off and a vacuum placed on the material for five minutes. The steam shall be reapplied for the time

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remaining in the autoclaving period and then a vacuum placed on the material for five minutes to remove all moisture Care must be taken to avoid the presence of creases in the cloth and any roll deformation since autoclaving will permanently set the cloth.

6 5 Only those chemical treatments already approved by the appropriate medical service and so listed in the invitation for bids or request for proposal shall be considered acceptable for the related procurement

* 6 6 Subject term (keyword) listing.

Antistatic
Aramid
Cloth
Herringbone twill
High temperature resistant
Twill

6 7 Changes from previous issue. The margins of this specification are marked with asterisks (or vertical lines) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-C-81280C	CLOTH, TWILL, ARAMID, STAPLE, HIGH TEMPERATURE RESISTANT

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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